

# 6DOF r13

## Schematic review

### UNTESTED PROTOTYPE - NOT A FUNCTIONAL TEST CERTIFICATE

This document preserves native EasyEDA schematic captures. Circuit symbols, wires and labels are not redrawn or inferred. Each image is fitted without cropping or stretching.

Identical net names are electrically connected, including across section outlines and separate detail pages. Graphic section boxes are visual organization, not electrical isolation barriers.

Overview pages are navigation aids. Use the detailed capture pages for pin names, component values and named-net review. Page order follows section numbers; the native source coordinate direction is retained.

Some component value attributes are hidden in the native schematic. Use the accompanying released BOM for exact component values, manufacturer part numbers and supplier identities; this PDF does not add missing values or infer them.

Firmware is deferred. USB-only power is for programming with drive outputs inhibited. This prototype is not a certified E-stop, STO or brake controller. Verify sequencing, thermal behavior, cable pinouts and operation before connecting powered servo motion.

### Section index

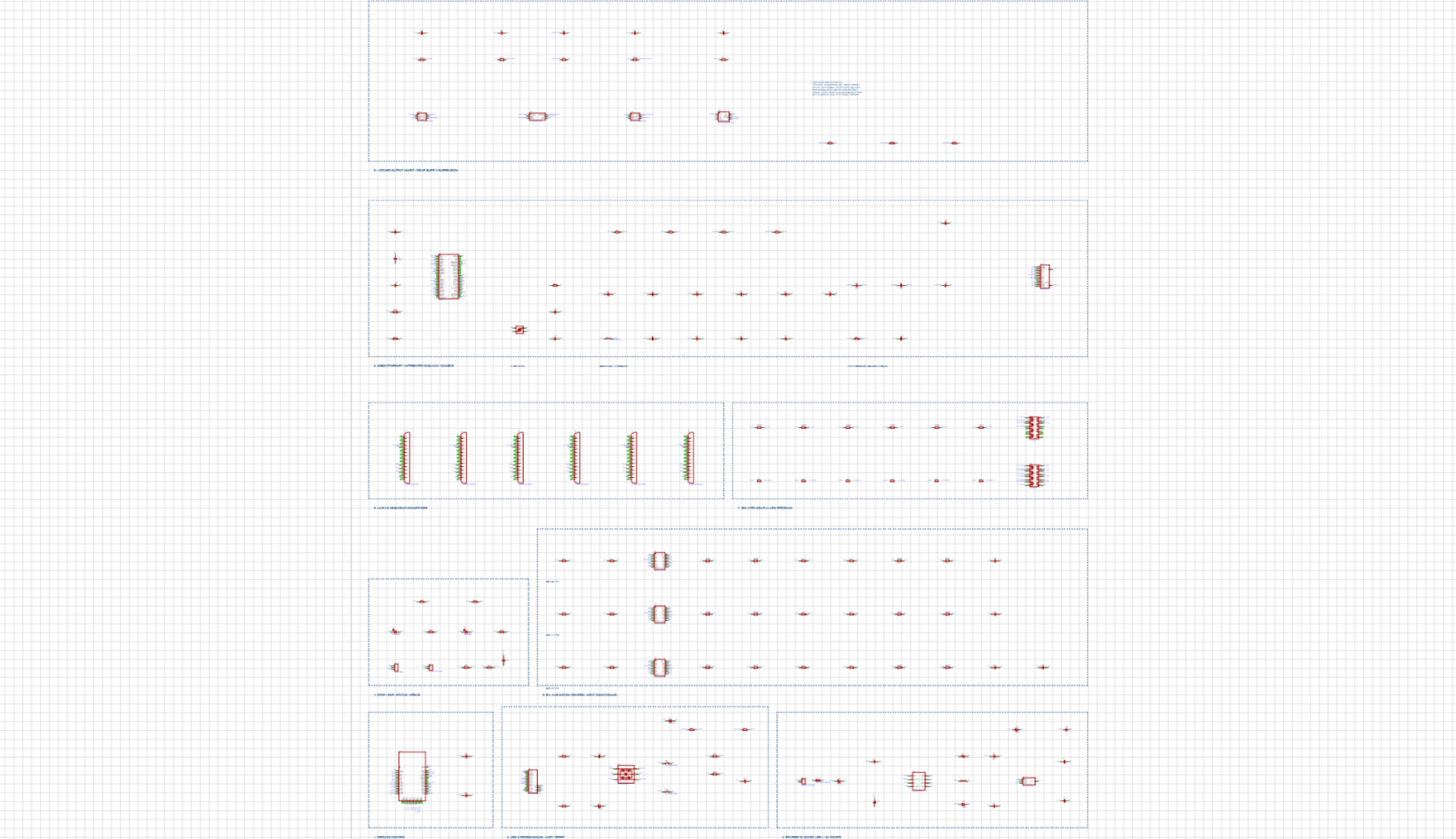
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| 2. USB-C PROGRAMMING / UART / RESET                  | page 4  |
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Schematic source SHA-256: 8f77a89e401a212a5dfa327bf64e8802e6e39c0c91044b6fa49c66b74c3ae1cf

Native capture date: 2026-09-23T18:16:39.998Z | A3 landscape. For print, use actual size and enlarge details as needed; do not shrink to A4.

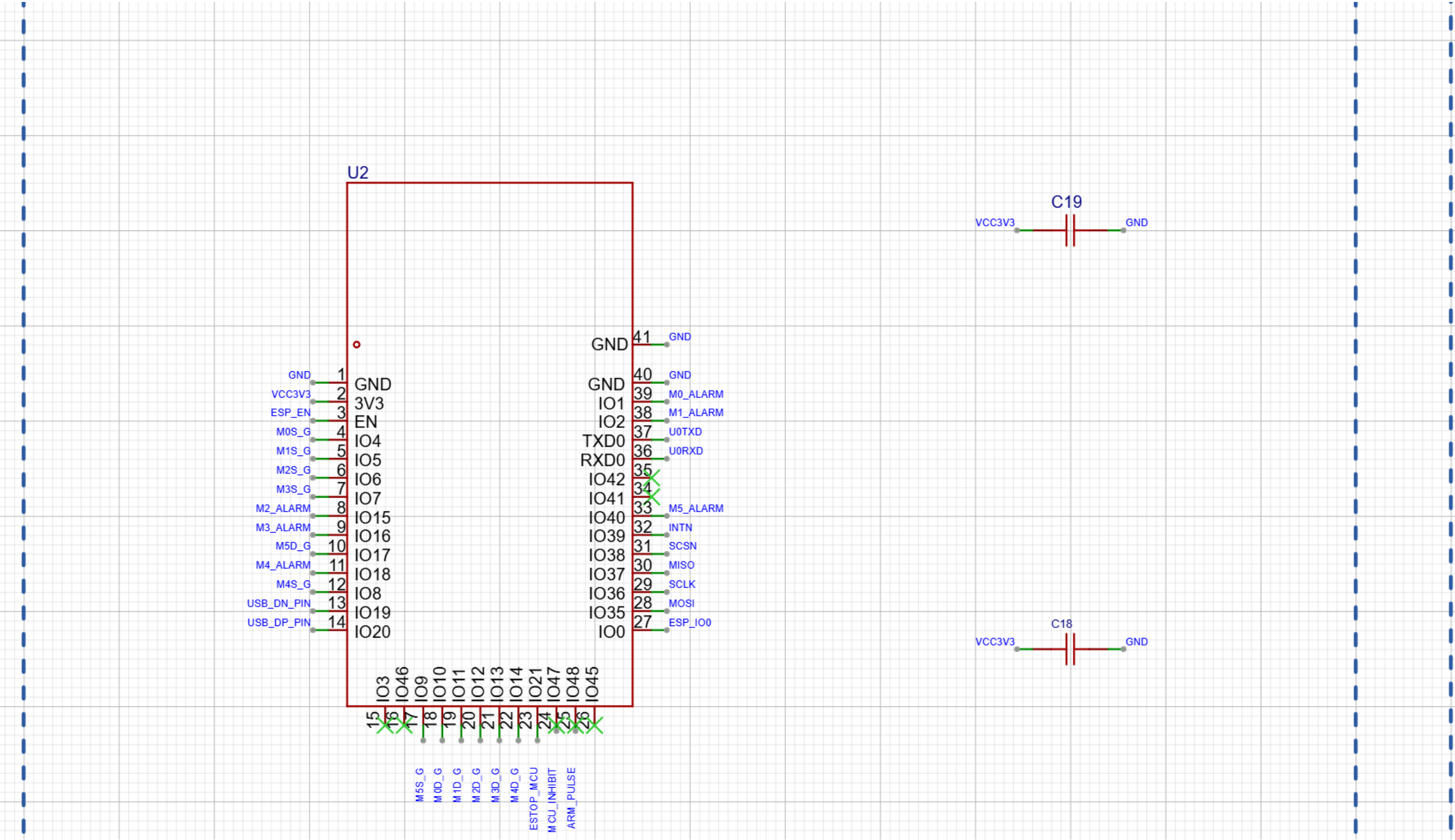
# Nine-section schematic overview

Navigation only. Section details follow; identical named nets connect across sections.



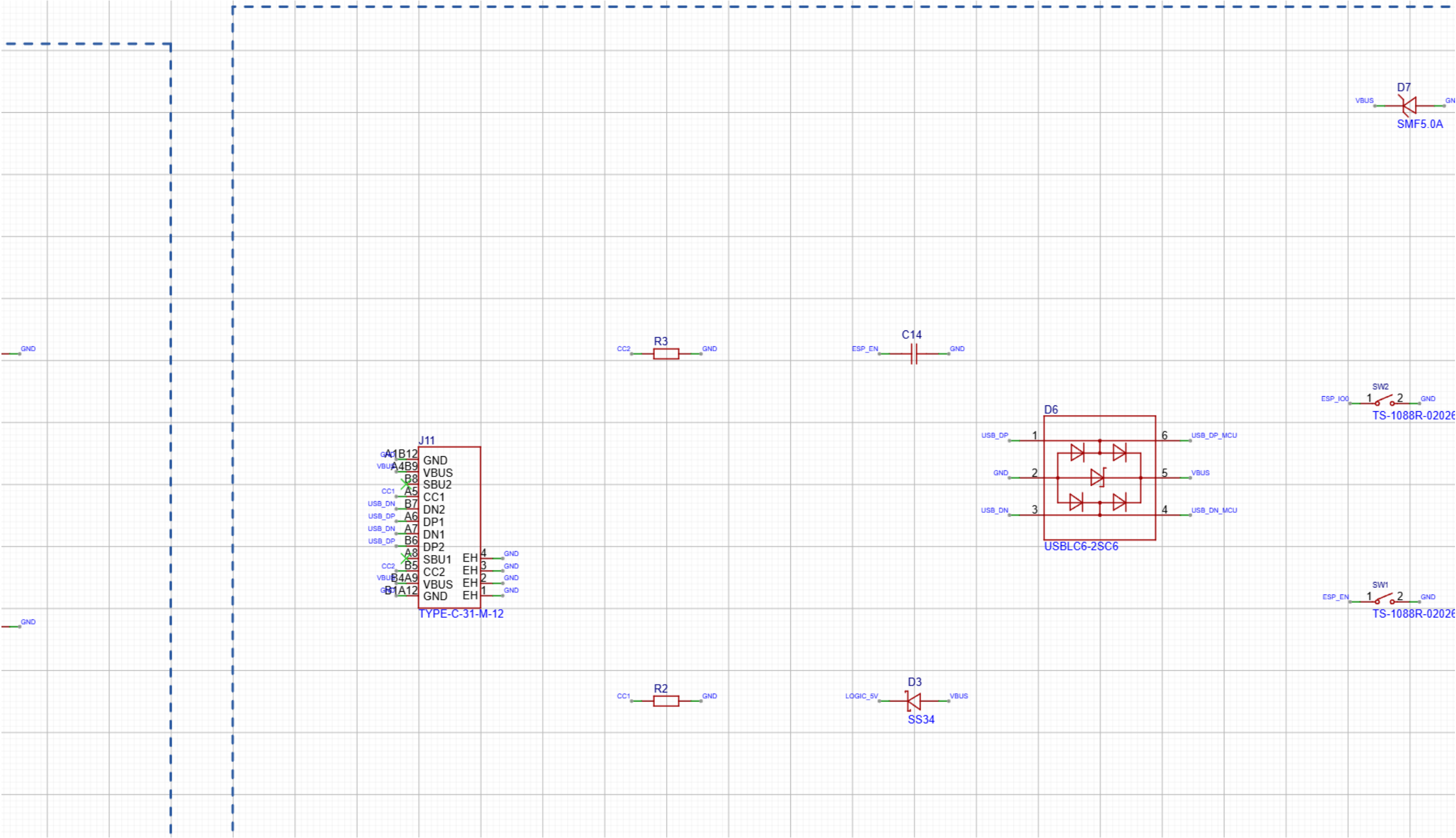
1. ESP32-S3 control

Unchanged native schematic detail. Overlap with adjacent views is intentional; identical named nets connect across pages.



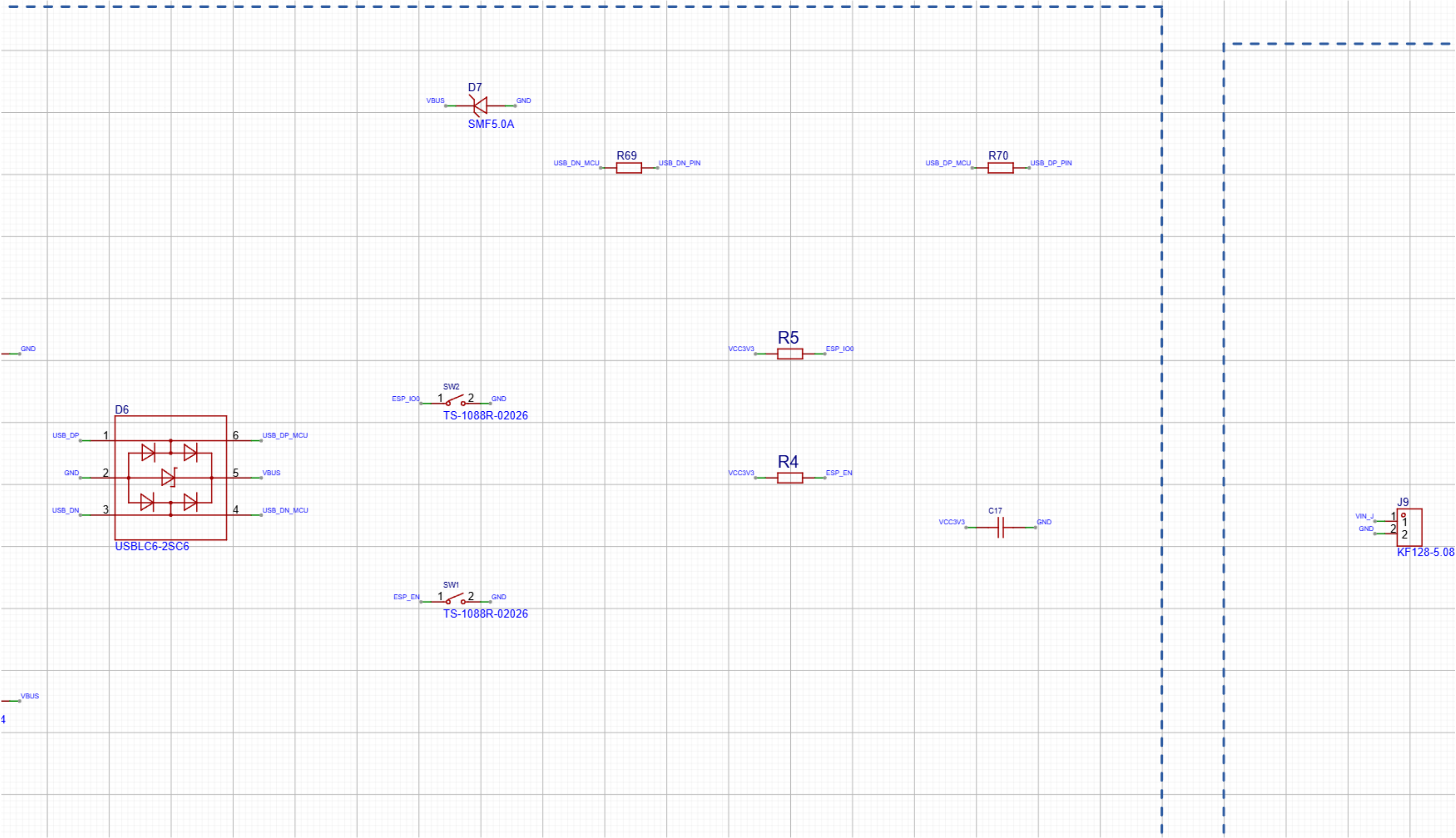
## 2. USB-C, power blocking and protection

Unchanged native schematic detail. Overlap with adjacent views is intentional; identical named nets connect across pages.



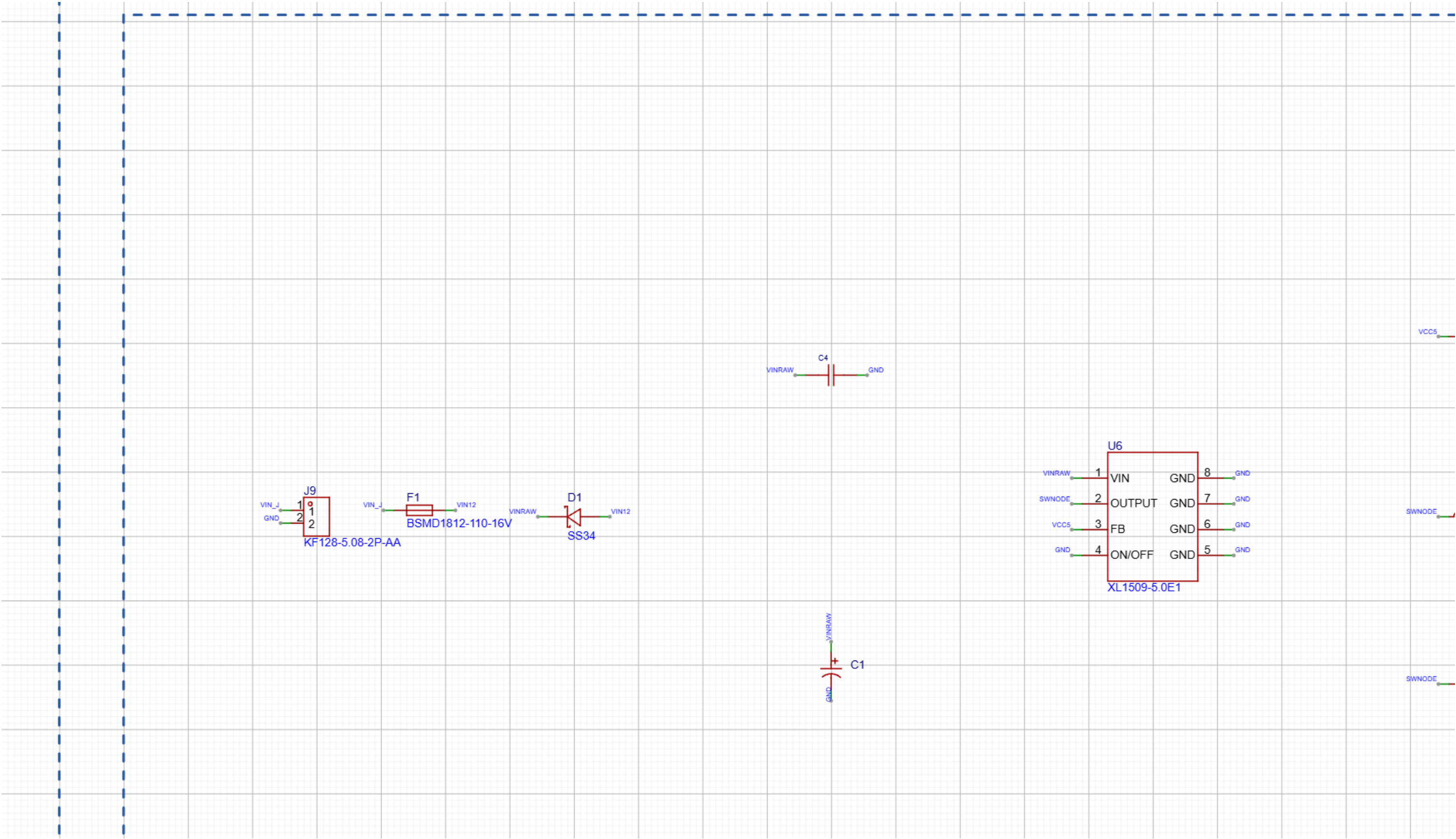
2. BOOT, RESET and USB series resistors

Unchanged native schematic detail. Overlap with adjacent views is intentional; identical named nets connect across pages.



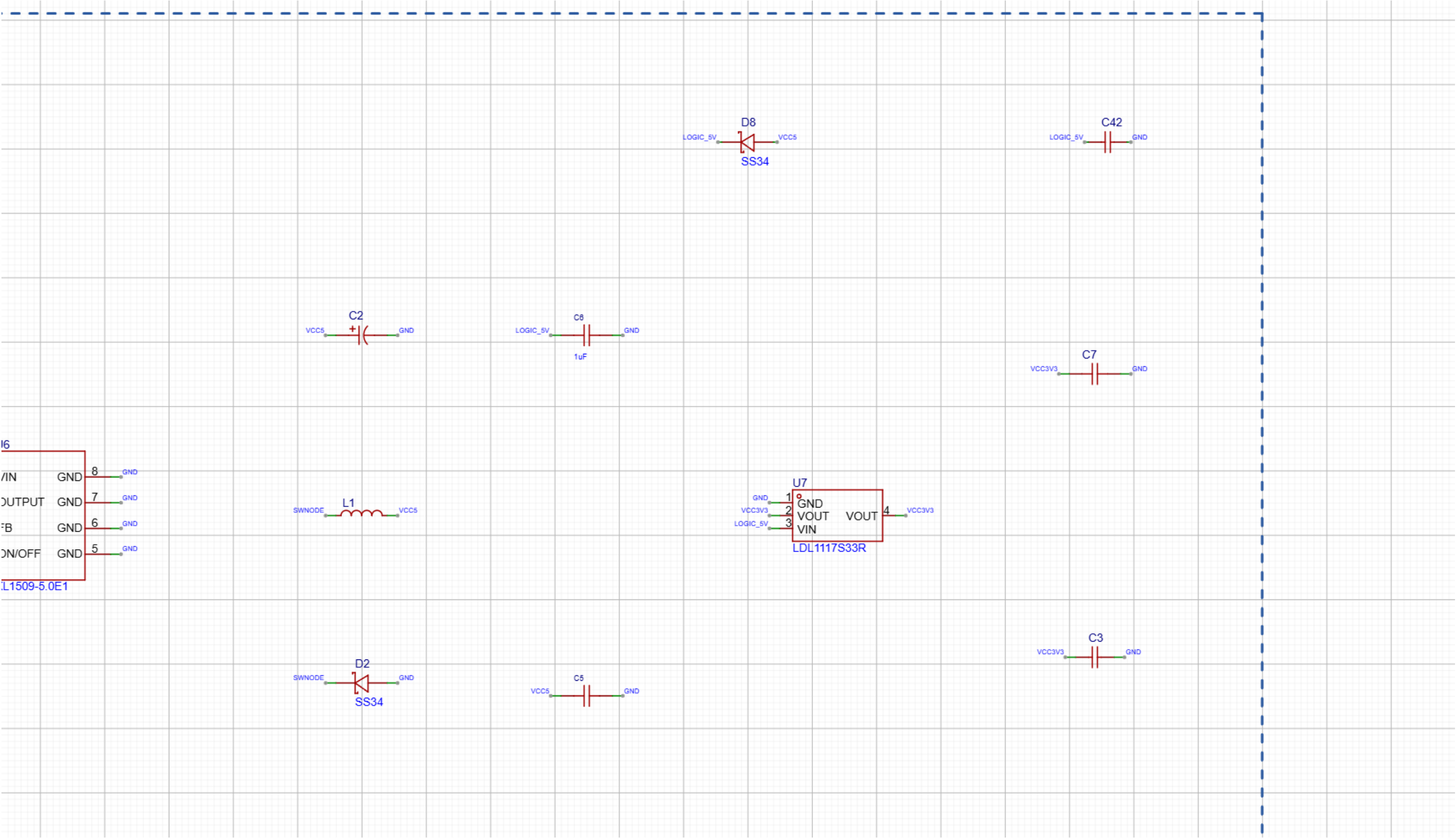
### 3. 12 V input protection and buck regulator

Unchanged native schematic detail. Overlap with adjacent views is intentional; identical named nets connect across pages.



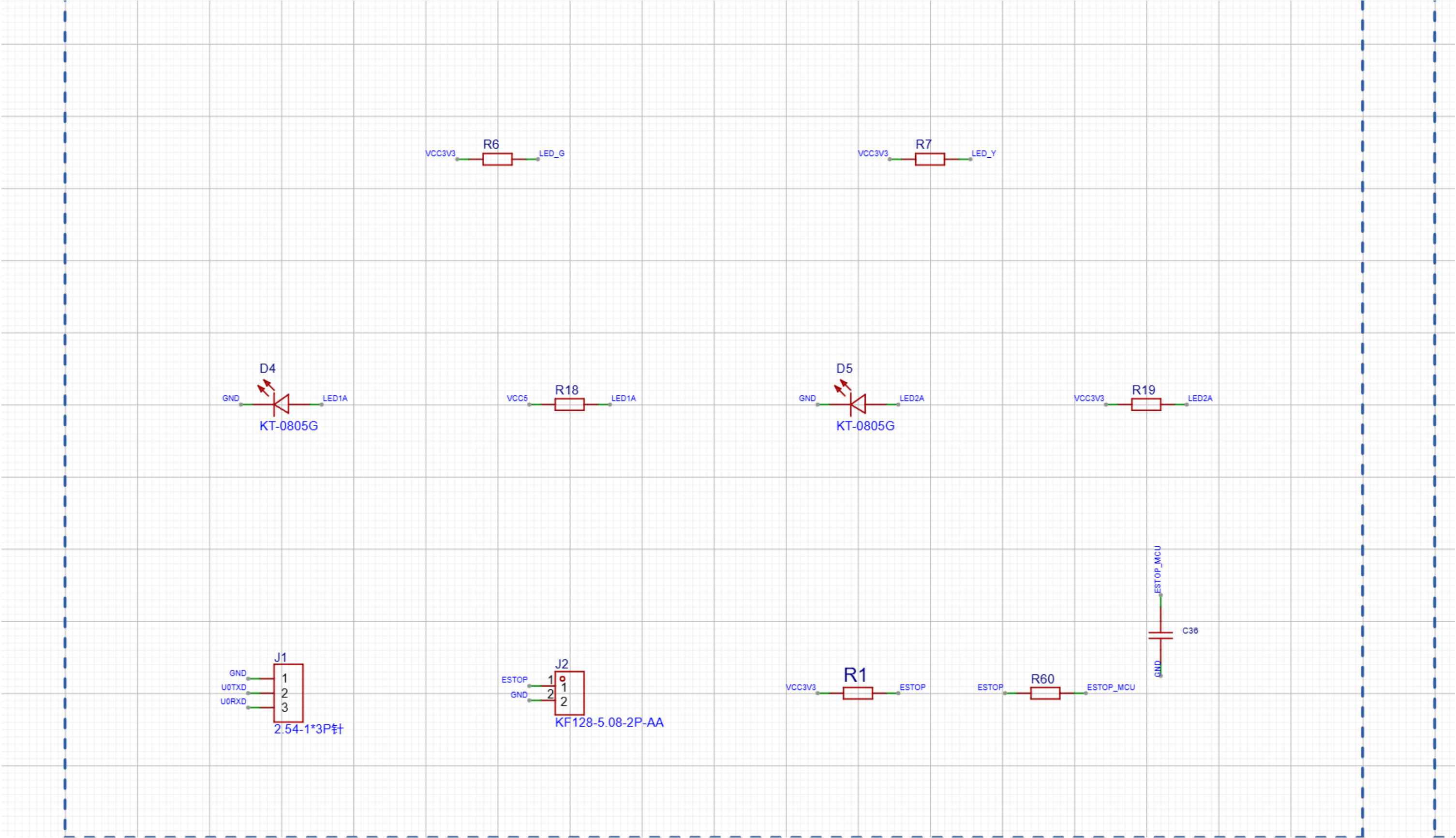
### 3. 5 V filtering, logic power OR and 3.3 V regulator

Unchanged native schematic detail. Overlap with adjacent views is intentional; identical named nets connect across pages.



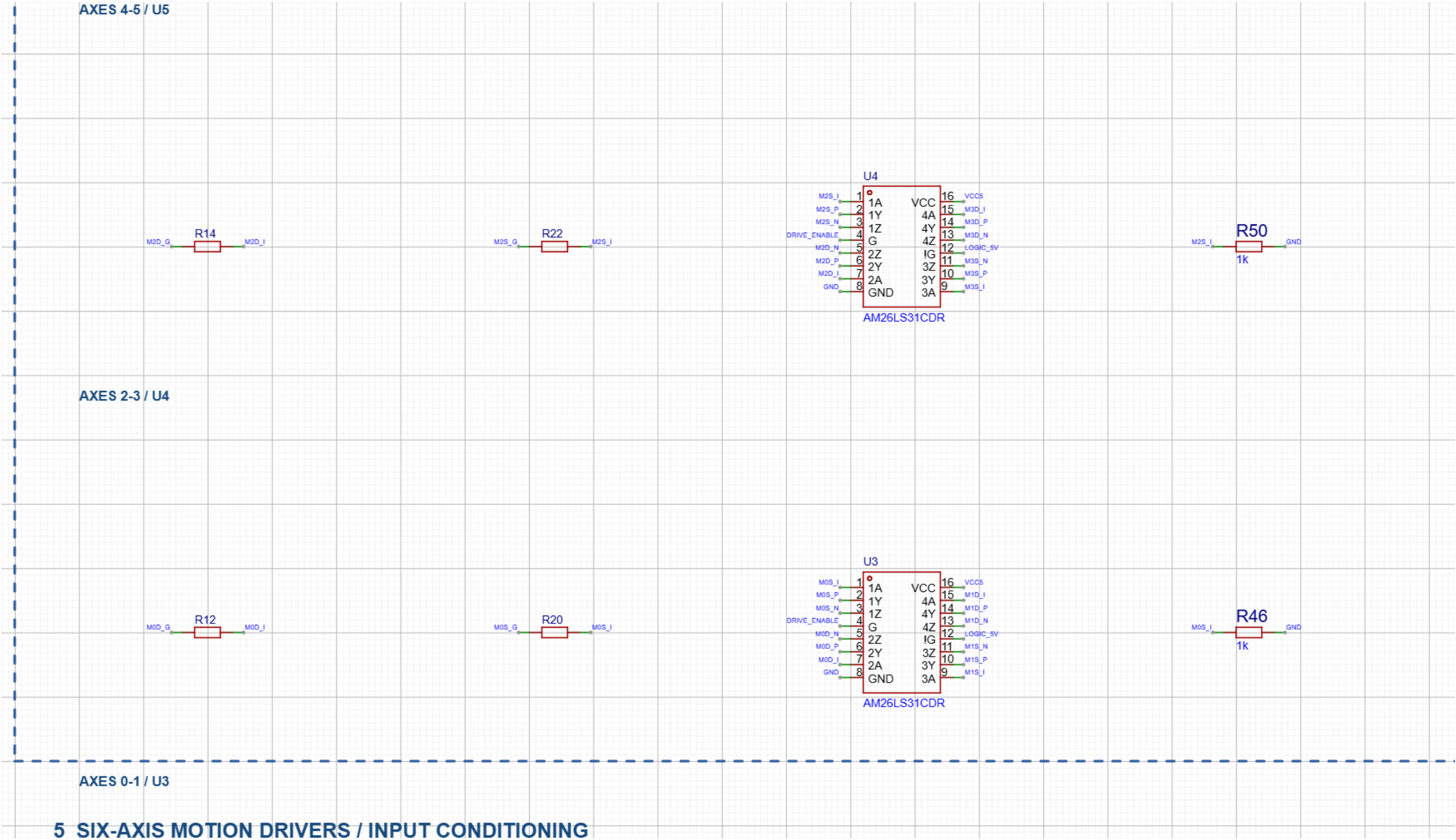
4. Stop loop, status and debug

Unchanged native schematic detail. Overlap with adjacent views is intentional; identical named nets connect across pages.



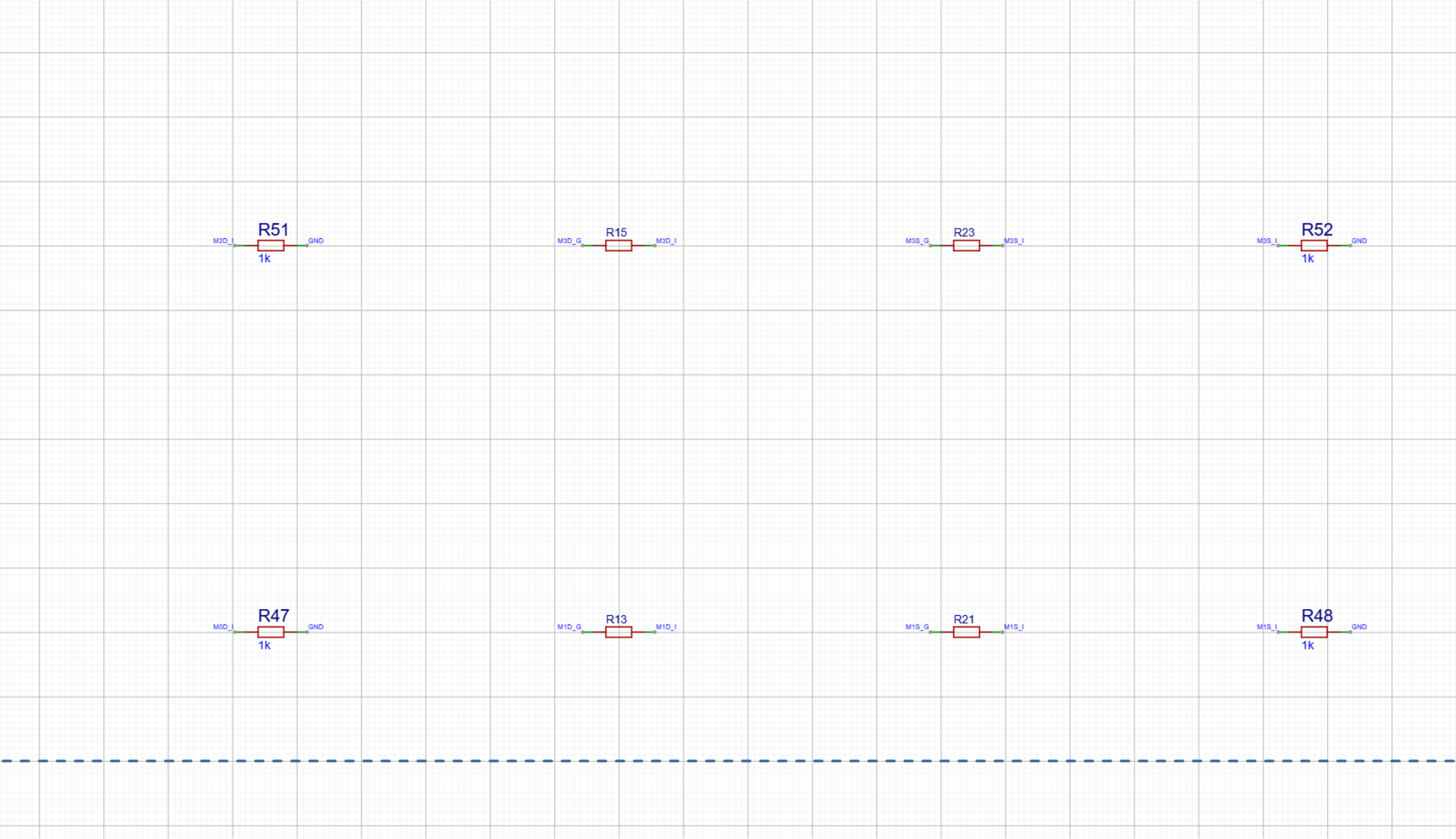
5. Motion drivers - left, axes 0-3

Unchanged native schematic detail. Overlap with adjacent views is intentional; identical named nets connect across pages.



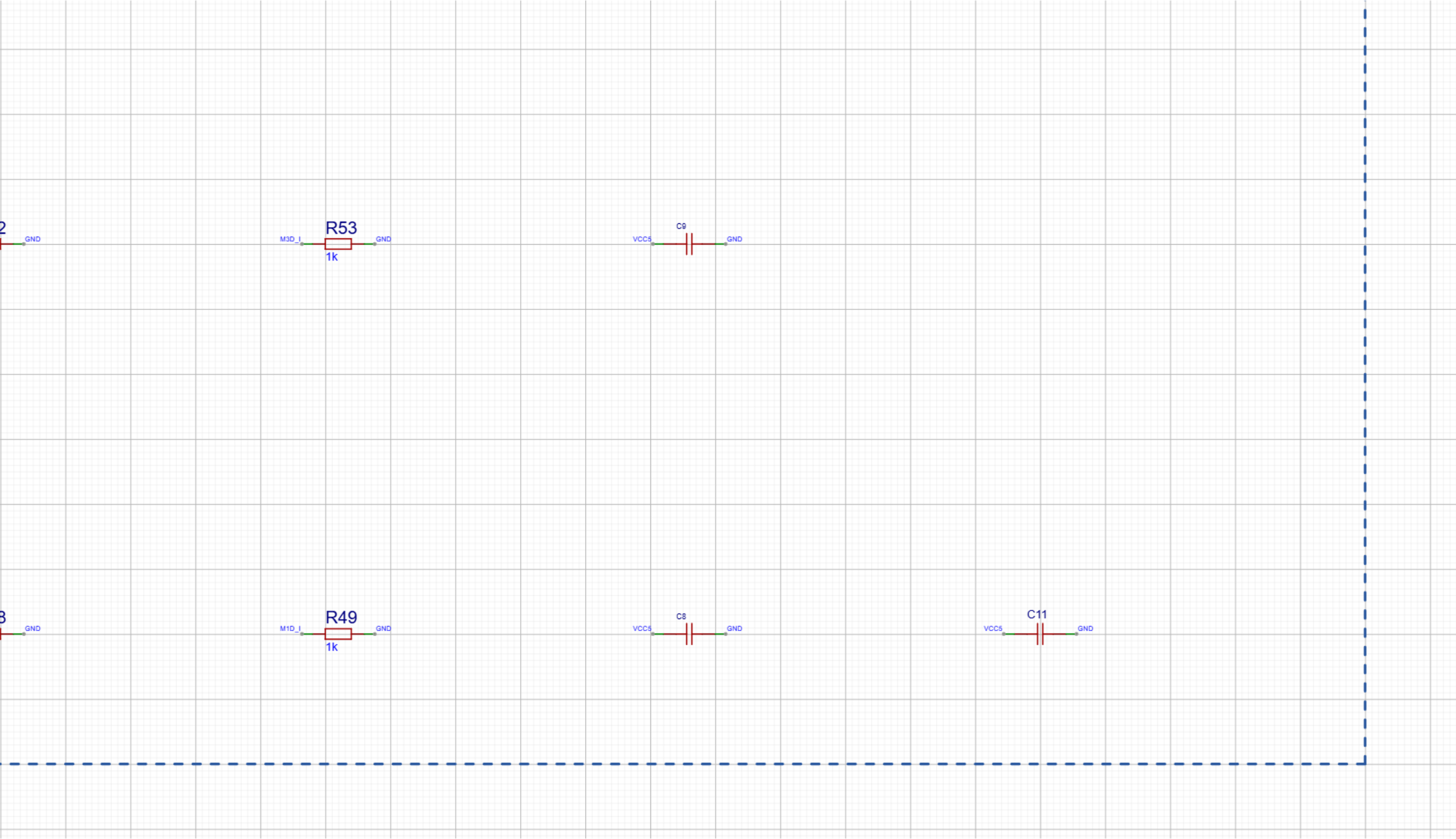
5. Motion drivers - middle, axes 0-3

Unchanged native schematic detail. Overlap with adjacent views is intentional; identical named nets connect across pages.



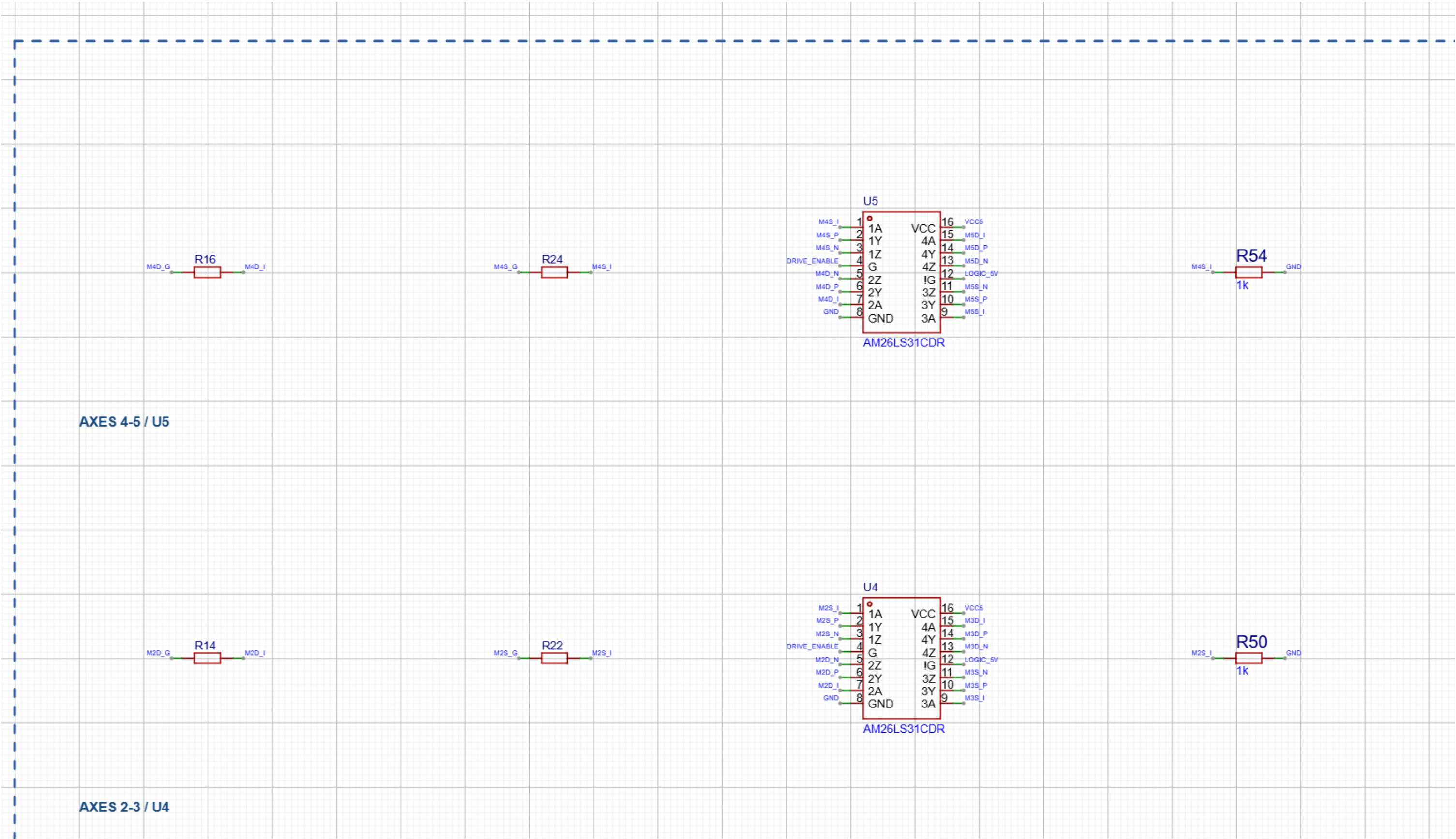
5. Motion drivers - right, axes 0-3

Unchanged native schematic detail. Overlap with adjacent views is intentional; identical named nets connect across pages.



5. Motion drivers - left, axes 2-5

Unchanged native schematic detail. Overlap with adjacent views is intentional; identical named nets connect across pages.



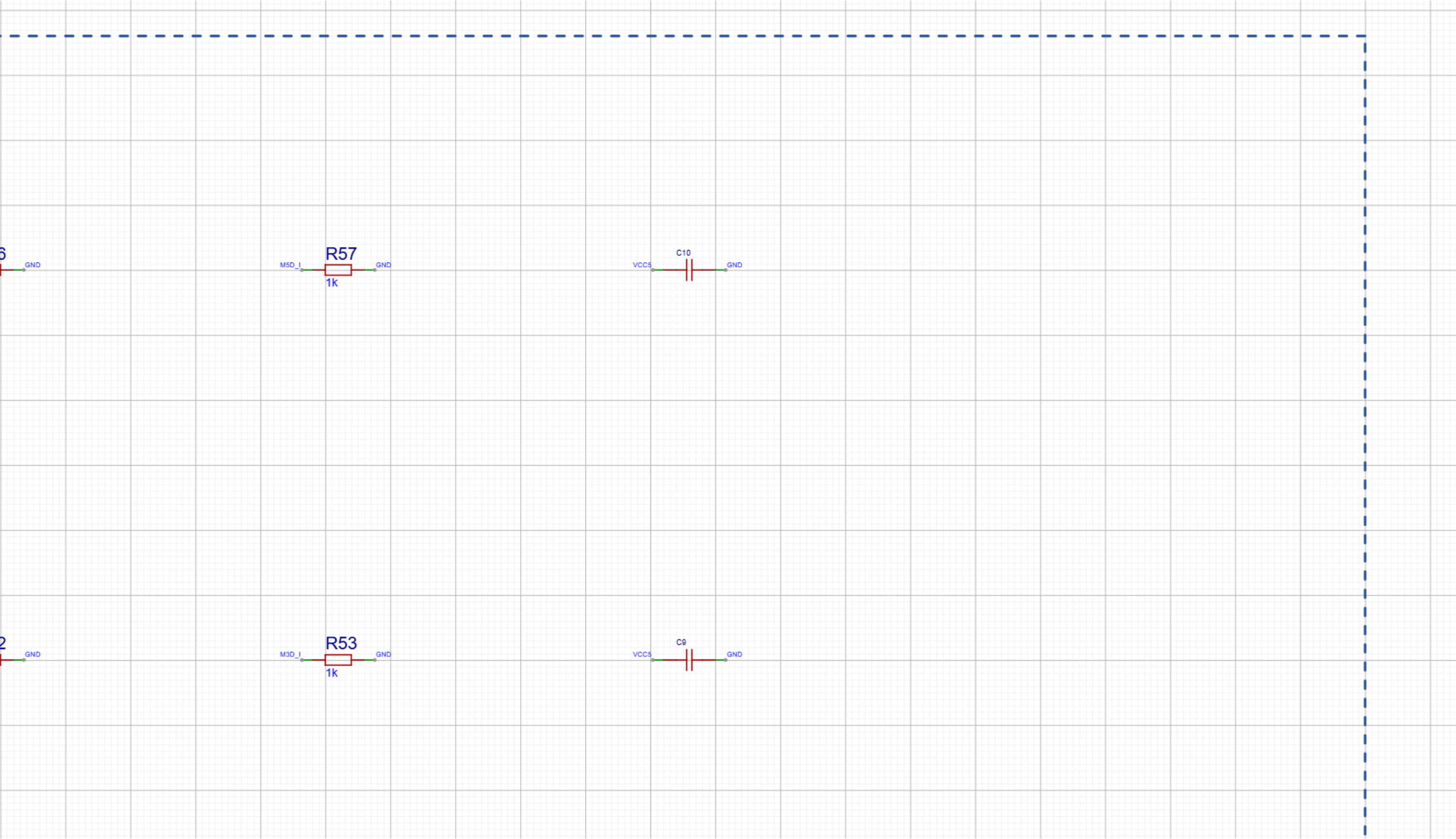
## 5. Motion drivers - middle, axes 2-5

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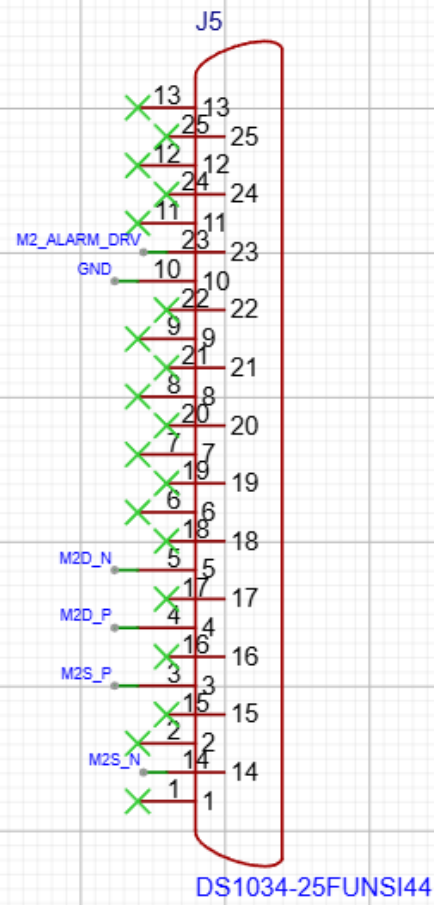
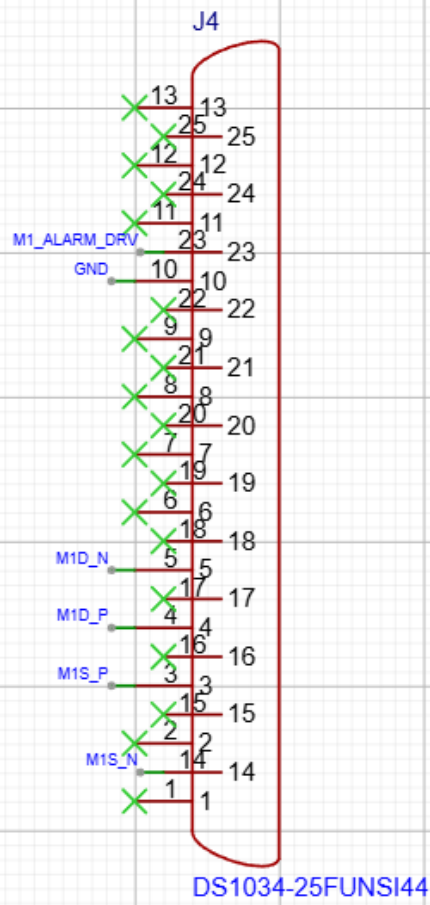
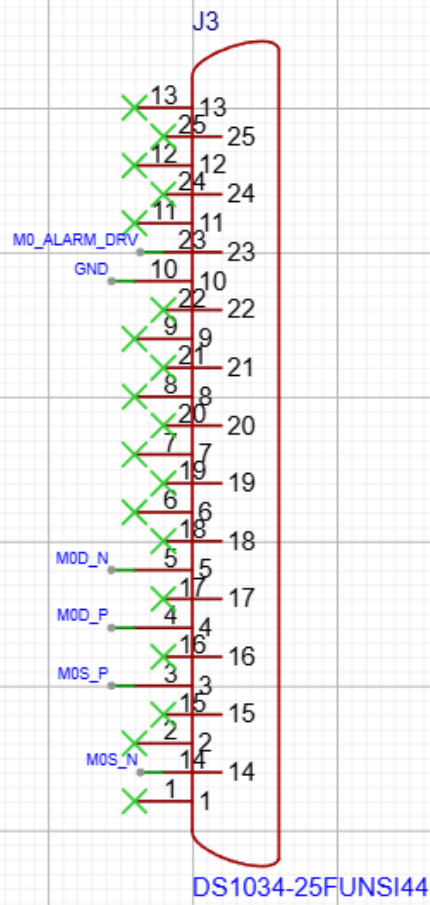
5. Motion drivers - right, axes 2-5

Unchanged native schematic detail. Overlap with adjacent views is intentional; identical named nets connect across pages.



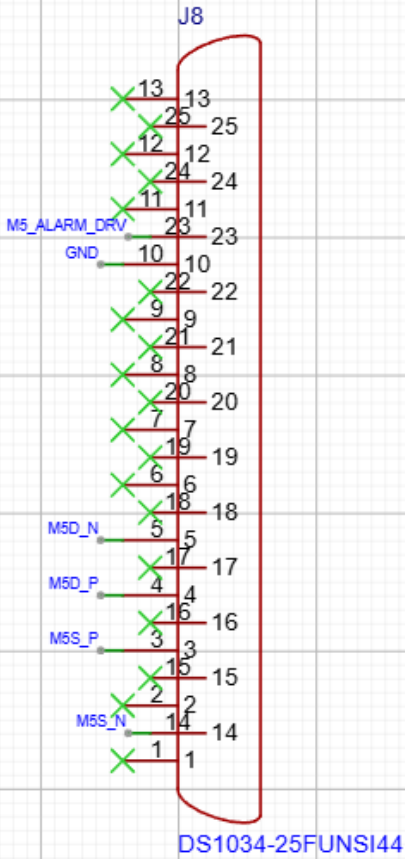
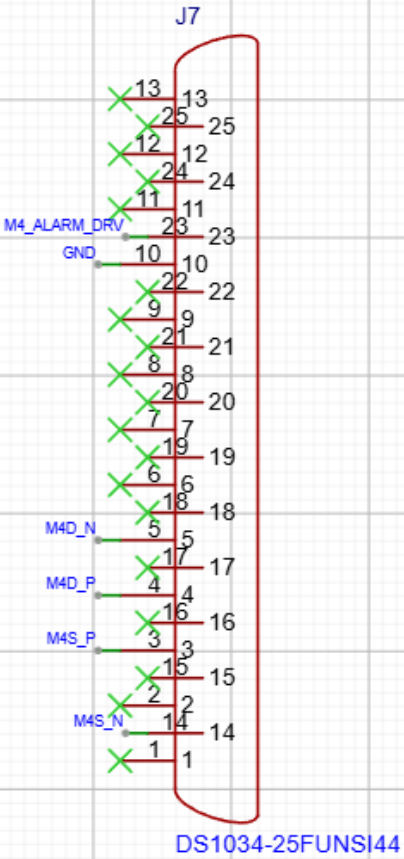
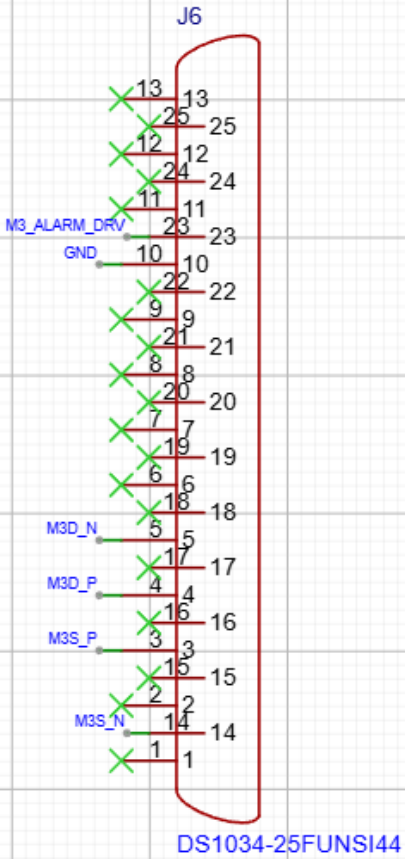
6. DB25 drive connectors - axes 0-2

Unchanged native schematic detail. Overlap with adjacent views is intentional; identical named nets connect across pages.



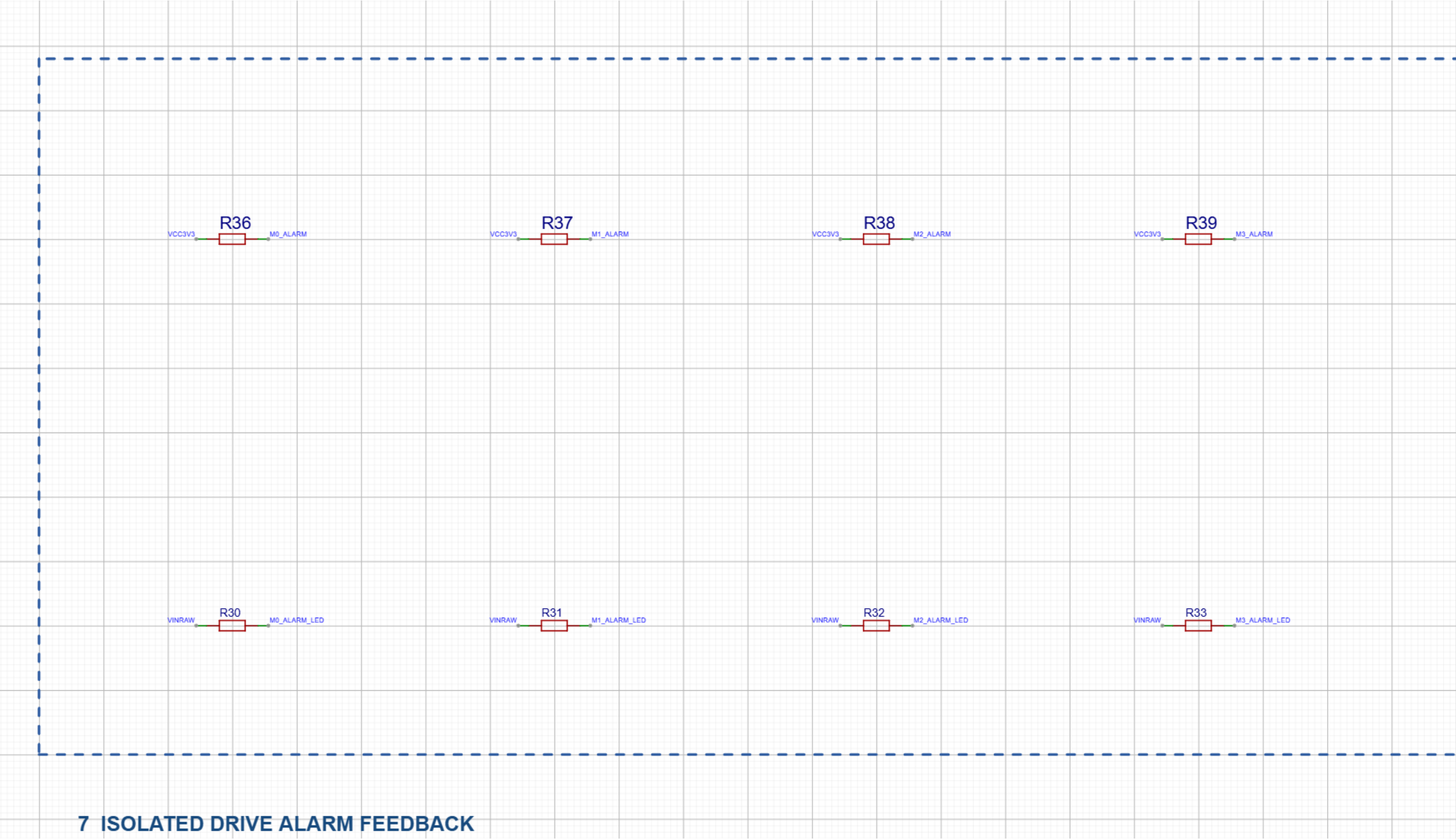
6. DB25 drive connectors - axes 3-5

Unchanged native schematic detail. Overlap with adjacent views is intentional; identical named nets connect across pages.



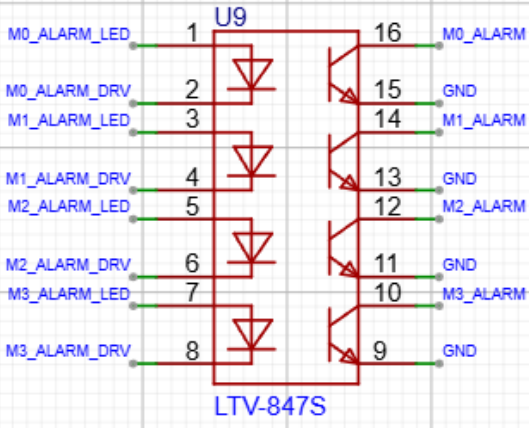
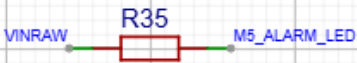
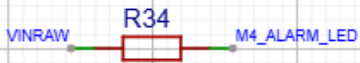
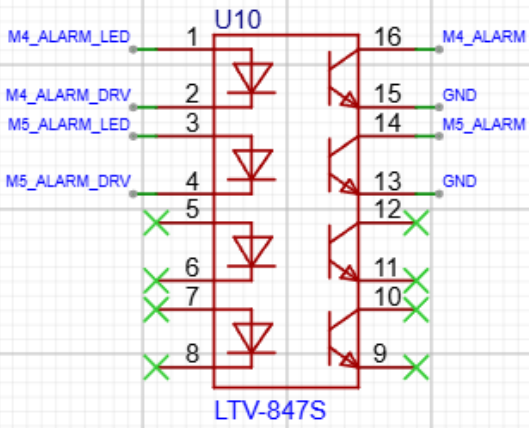
## 7. Drive alarm feedback - input networks

Unchanged native schematic detail. Overlap with adjacent views is intentional; identical named nets connect across pages.



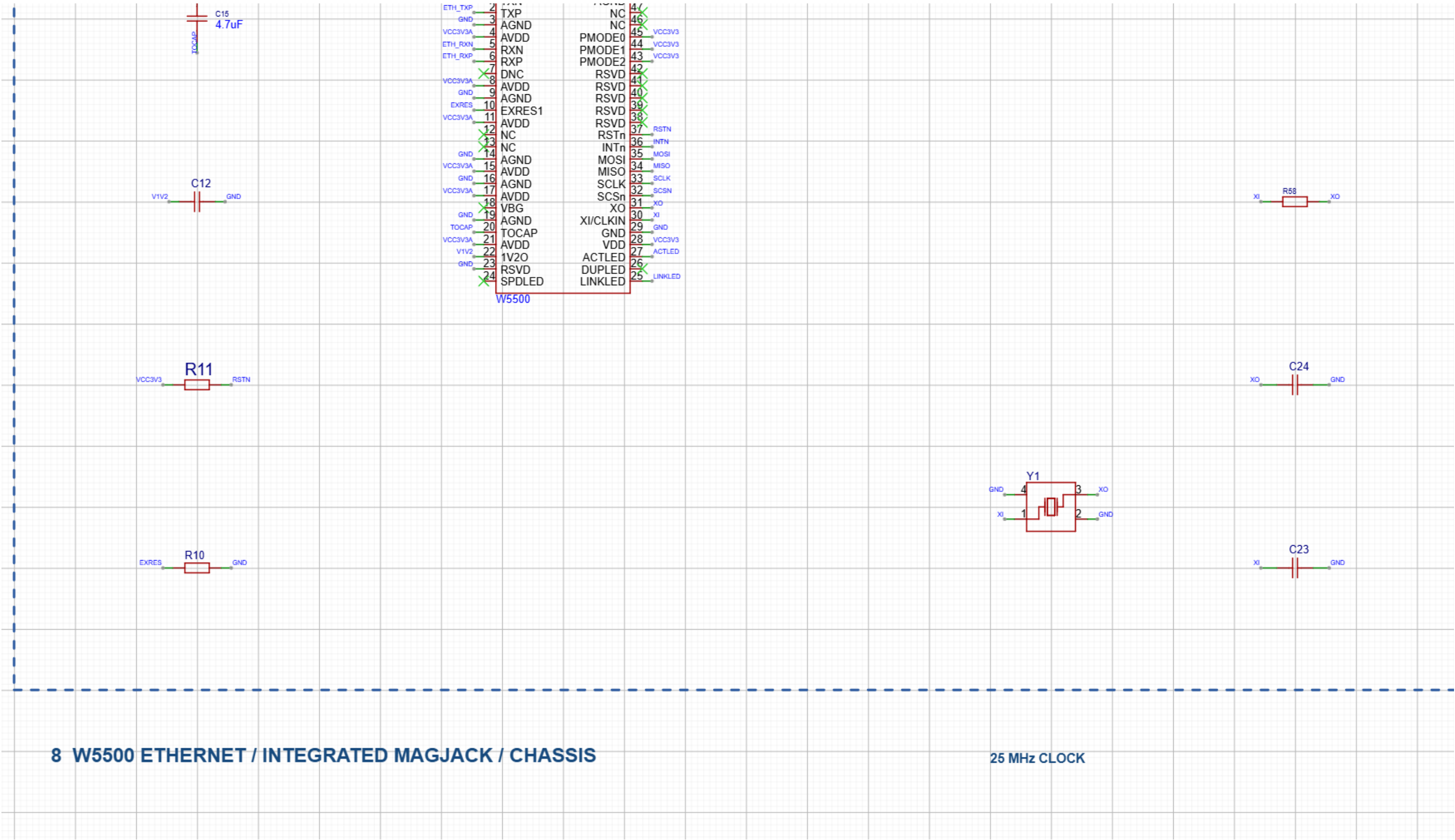
7. Drive alarm feedback - optocouplers

Unchanged native schematic detail. Overlap with adjacent views is intentional; identical named nets connect across pages.



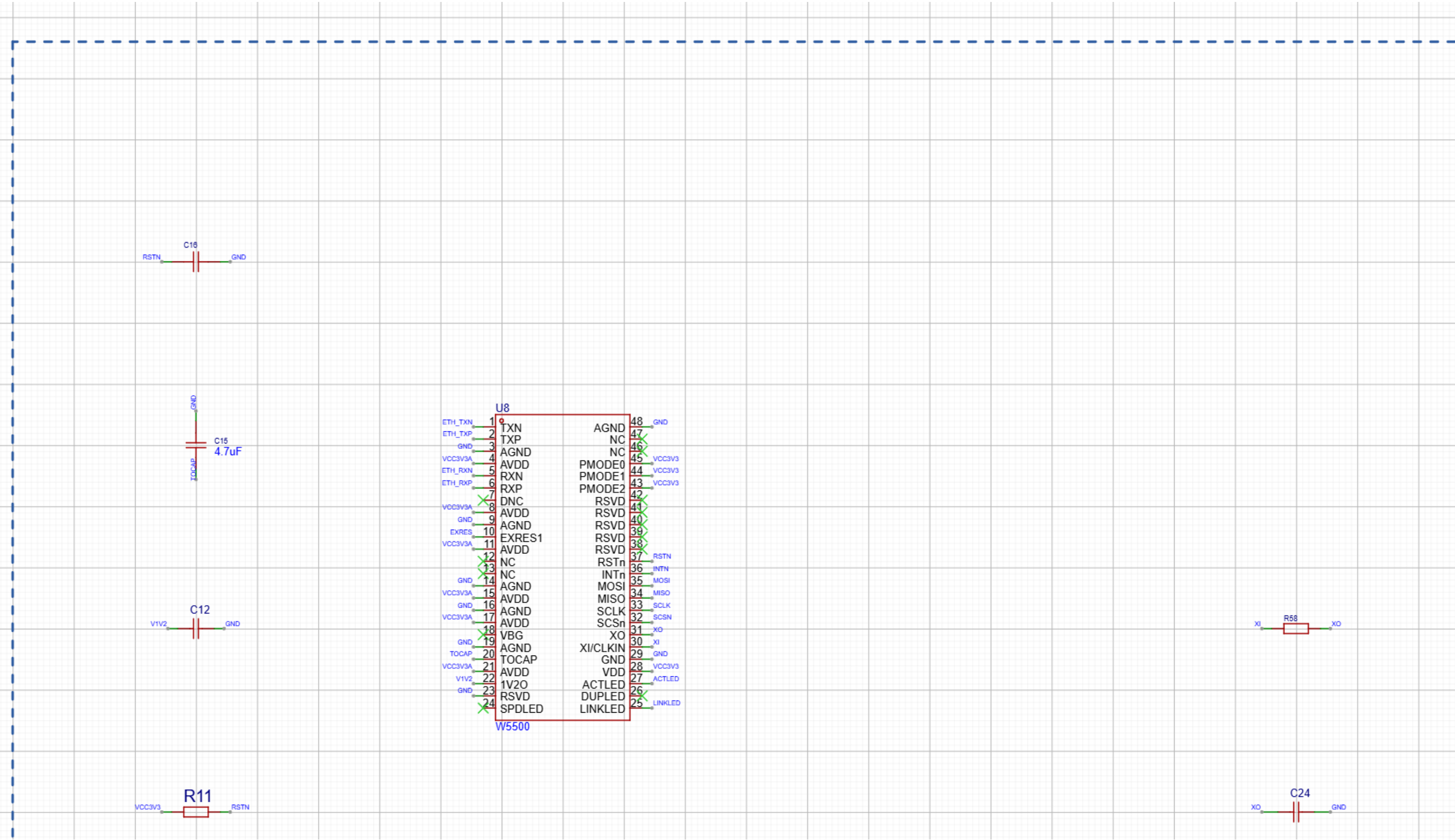
8. Ethernet - W5500 upper pins and clock

Unchanged native schematic detail. Overlap with adjacent views is intentional; identical named nets connect across pages.



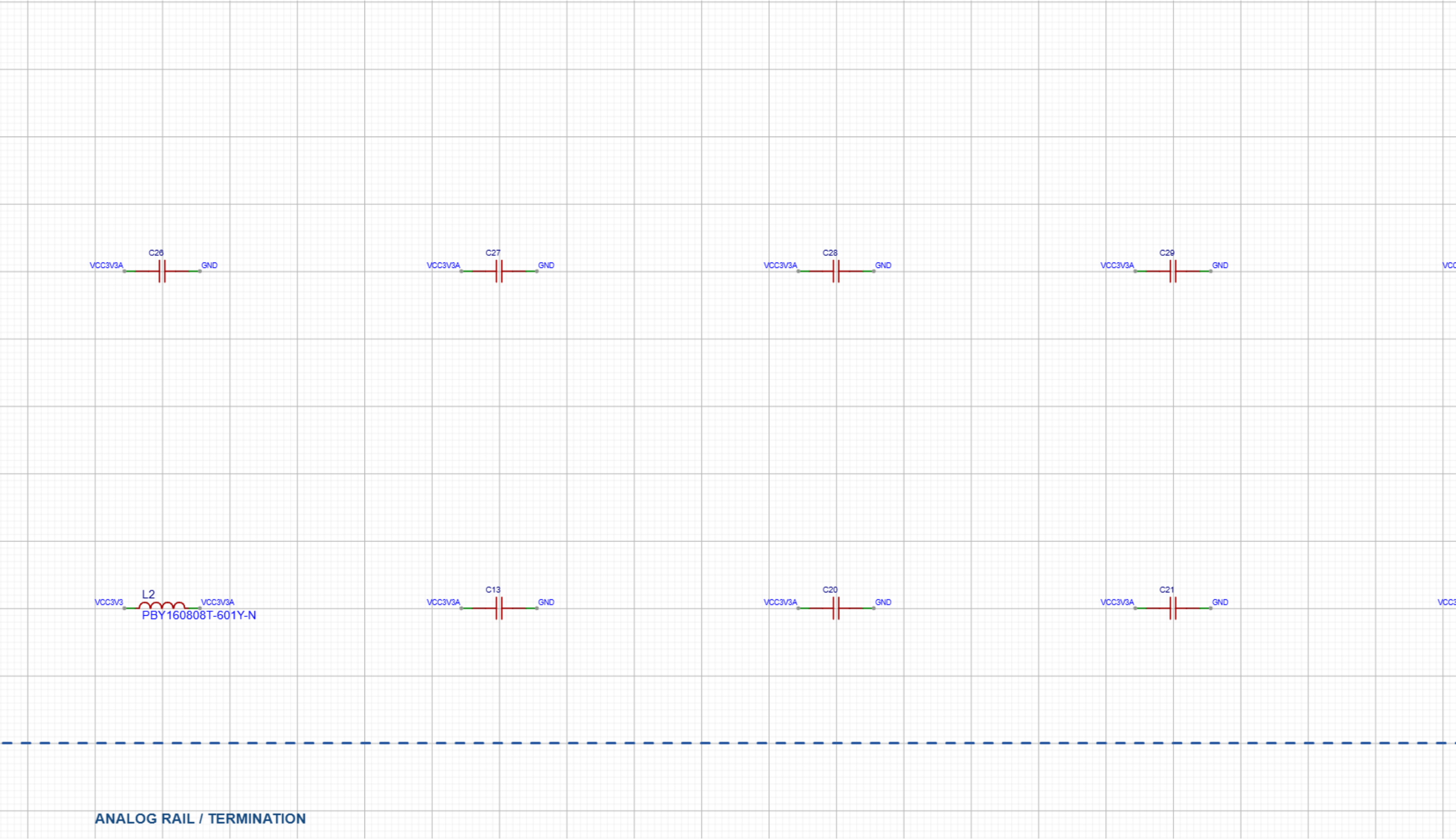
8. Ethernet - W5500 lower pins and bypass

Unchanged native schematic detail. Overlap with adjacent views is intentional; identical named nets connect across pages.



## 8. Ethernet - analog supply and bypass, left

Unchanged native schematic detail. Overlap with adjacent views is intentional; identical named nets connect across pages.



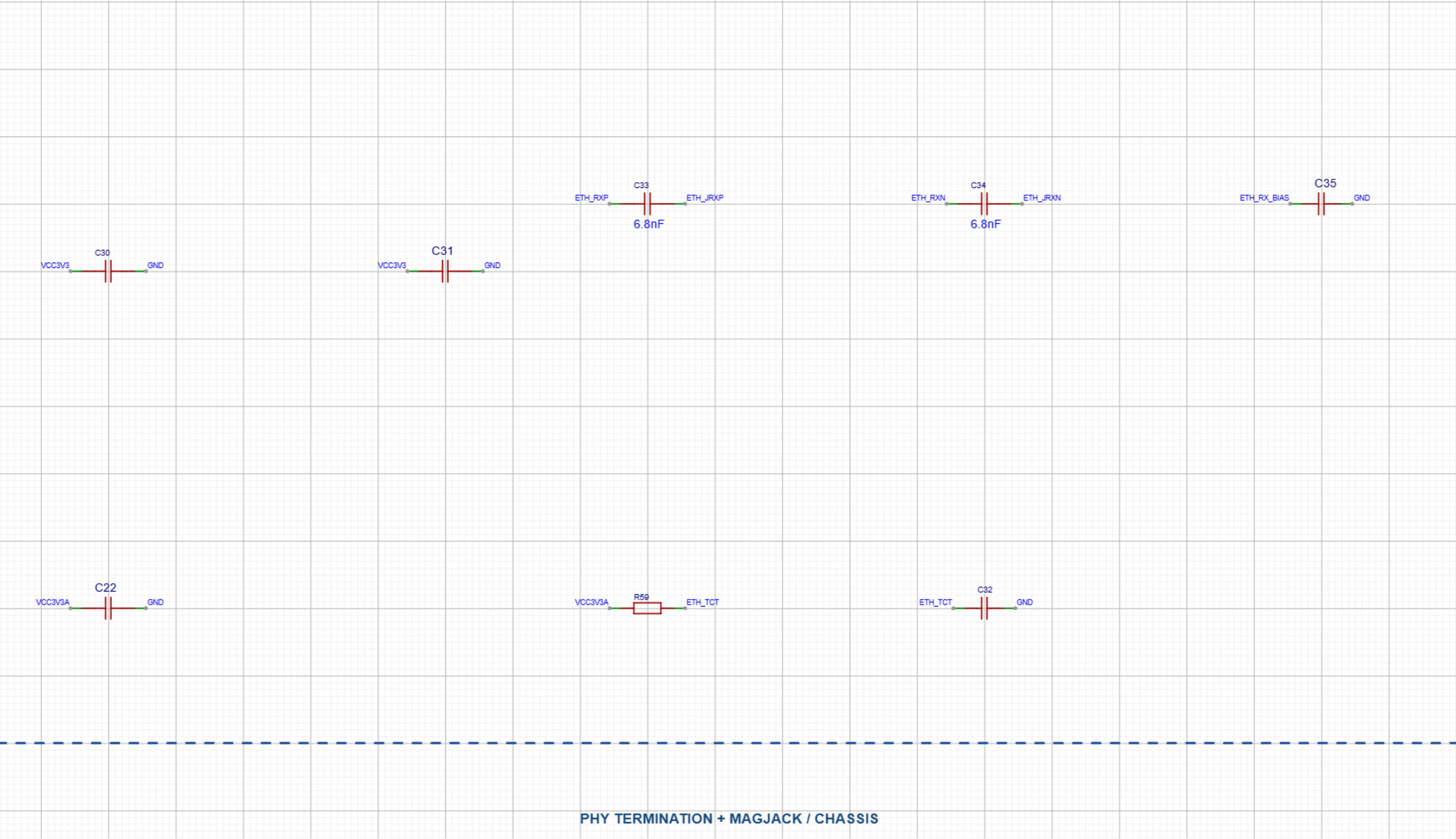
8. Ethernet - termination, left

Unchanged native schematic detail. Overlap with adjacent views is intentional; identical named nets connect across pages.



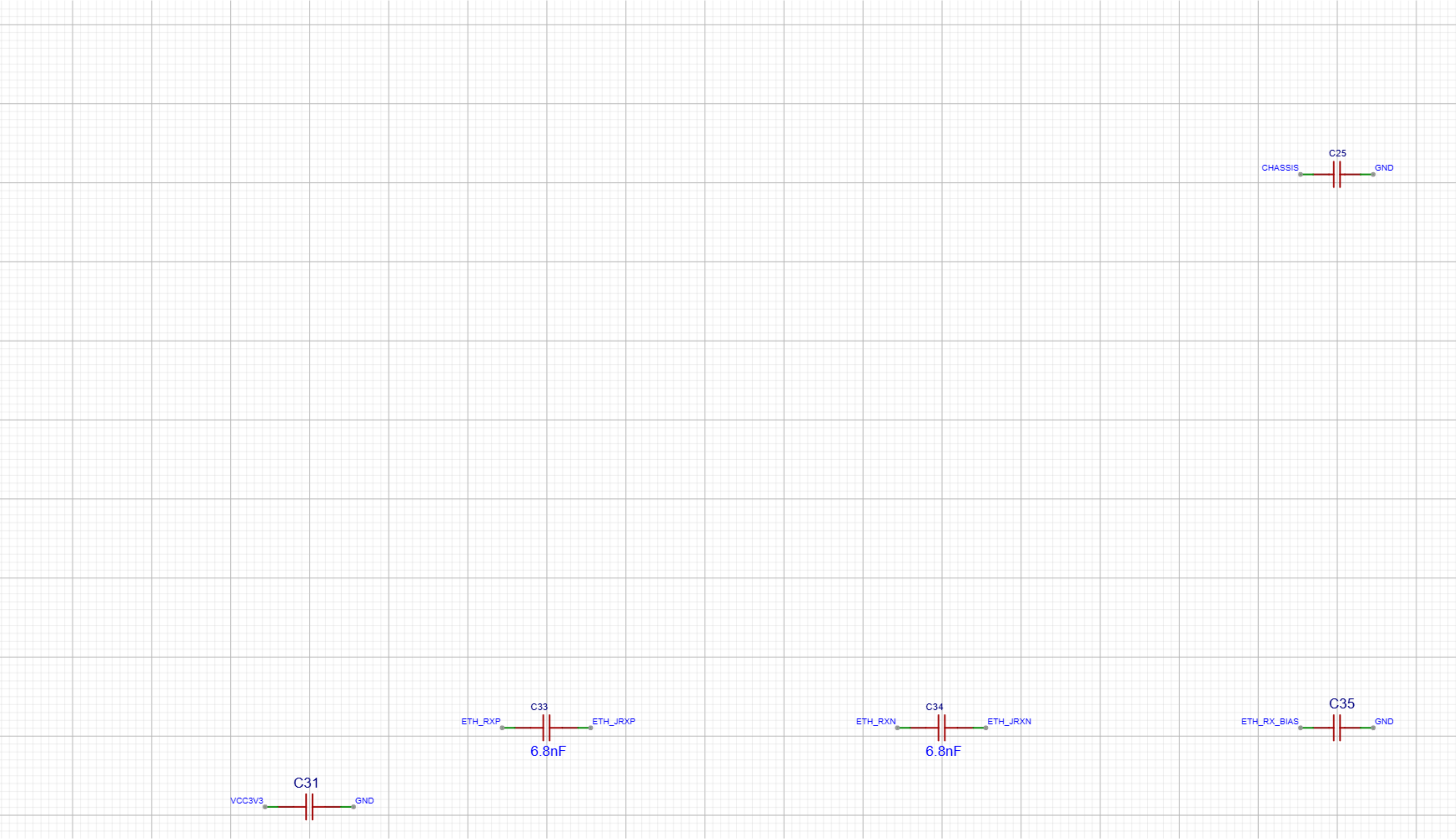
## 8. Ethernet - analog supply and bypass, right

Unchanged native schematic detail. Overlap with adjacent views is intentional; identical named nets connect across pages.



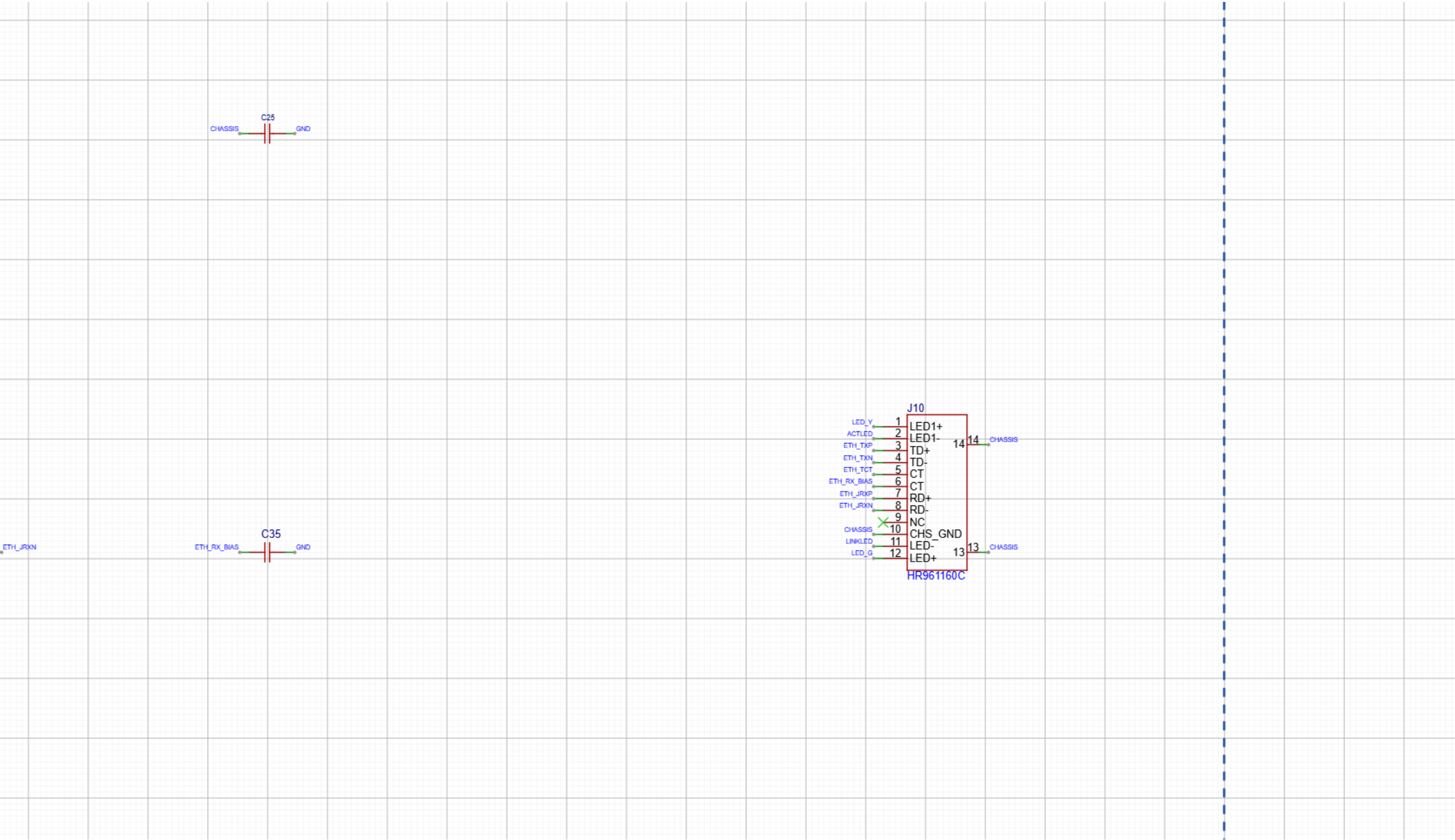
## 8. Ethernet - chassis coupling network

Unchanged native schematic detail. Overlap with adjacent views is intentional; identical named nets connect across pages.



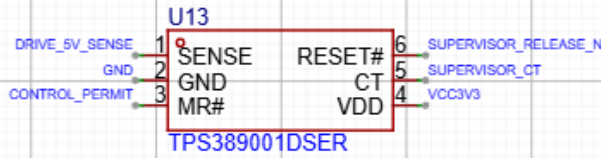
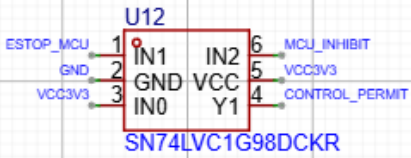
## 8. Ethernet - integrated MagJack

Unchanged native schematic detail. Overlap with adjacent views is intentional; identical named nets connect across pages.



## 9. Output inhibit - input logic and supervisor

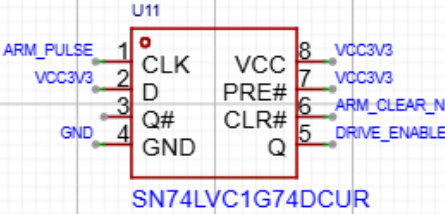
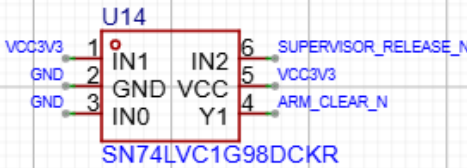
Unchanged native schematic detail. Overlap with adjacent views is intentional; identical named nets connect across pages.



### 9 LATCHED OUTPUT INHIBIT / DRIVE-SUPPLY SUPERVISION

## 9. Output inhibit - clear buffer and arm latch

Unchanged native schematic detail. Overlap with adjacent views is intentional; identical named nets connect across pages.



## 9. Output inhibit - input bias and supervisor timing

Unchanged native schematic detail. Overlap with adjacent views is intentional; identical named nets connect across pages.



## 9. Output inhibit - latch bias and bypass

Unchanged native schematic detail. Overlap with adjacent views is intentional; identical named nets connect across pages.



## 9. Driver-enable bias and prototype operating notes

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UNTESTED PROTOTYPE r13  
USB alone: programming only; outputs inhibited.  
GPIO47 HIGH inhibits. GPIO48 fresh edge arms.  
Stop opening / driver-rail loss clears the latch.  
Closure / power return never automatically re-arms.  
NOT a certified E-stop, STO or brake controller.

